Steering a Network with Its Own Reflection: True, False, and Noise Self-Model Broadcasts in an Evolving Multi-Agent Network

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Abstract

Many real networks respond to measurements of their own state: markets react to indices, organizations to metrics, online platforms to trending signals. We introduce a controlled laboratory for this reflexive loop. A population of temporary agents harvests resources, cooperates, reproduces with mutation, and dies, collectively forming an adaptive network $G(t)$. A global observer periodically compresses the macrostate (fragmentation, centralization, cooperation, inequality, turnover) into a self-model $S(t)$ that is broadcast back to the agents either accurately, systematically distorted, or replaced by matched-bandwidth noise; two further conditions withhold the broadcast, one while still measuring. Across 970 deterministic, seed-replayable runs with paired seeds and pre-registered outcomes, we find: (i) measurement without broadcast is causally inert (outcome distributions identical to baseline, Cliff's $\delta = 0$); (ii) a systematically false self-model pulls the macrostate toward itself far more strongly than noise of identical bandwidth (δ up to 0.85, $p \approx 10^{-8}$), with the effect largest, relative to noise, at low feedback gain; (iii) the direction of the effect depends on the type of distortion — an inverted self-model exploits corrective responses, a permanent-crisis self-model is self-defeating yet maximizes cooperation, and a flattering self-model is behaviorally inert; and (iv) heritable sensitivity to the broadcast is selected against in proportion to its unreliability and volume: populations fed false self-models evolve to ignore their observer within tens of generations. Accurate feedback roughly doubles costly cooperation, while post-shock recovery time is insensitive to feedback in all conditions — a robust negative result. The framework offers a reproducible causal testbed for reflexivity in adaptive networks, from telemetry-driven infrastructure to measured social systems.

1 Introduction

Adaptive networks — graphs whose topology coevolves with the dynamical states of their nodes — are a central object of network science [1, 2]. A distinctive subclass has received far less controlled attention: networks that respond to *descriptions of themselves*. Financial markets move on published indices computed from their own trades [3]; universities restructure around rankings derived from their own behavior [4]; recommendation platforms amplify items *because* an internal measurement declared them trending. In each case a macroscopic measurement of the system is compressed, published, and re-enters the system as an input to microscopic decisions — a loop Merton called the self-fulfilling prophecy [5] and whose measure-distorting variant is known as Goodhart's law [6].

Empirical study of this loop is confounded: in real systems one can rarely withhold the measurement, falsify it in controlled directions, or replay history under identical randomness. We therefore built a minimal simulated world in which the loop can be opened, closed, and corrupted at will, and where every run is exactly reproducible from a configuration and seed. Agents are deliberately simple — transparent trait vectors rather than learned policies — so that every behavioral pathway from broadcast to response is inspectable.

Our contributions are: (1) an open, deterministic experimental platform coupling an evolving agent ecology, an adaptive interaction network, a global observer, and a controllable broadcast channel; (2) a five-condition causal design that separates the effect of *being measured* from the effect of *being told*, and the effect of *self-information* from that of *stimulation* (via a matched-bandwidth noise control); (3) evidence, robust across feedback gains and two shock severities, that false self-models steer the macrostate toward themselves while noise does not, and that the steering direction is a function of the distortion type; and (4) to our knowledge the first demonstration in an ecological agent-based setting that *attention to a collective self-model is itself an evolvable trait*, selected against in proportion to the self-model's unreliability.

2 Related Work

Adaptive and temporal networks. Coevolution of state and topology is surveyed in [1, 2]; our model belongs to this family, with births, deaths and rewiring driven by agent decisions. Robustness of networks to random and targeted node removal is classical [7]; we use targeted hub removal as a standardized perturbation rather than an object of study. **Global coupling in physics.** Globally coupled oscillators and maps show that a mean field fed back to units reshapes collective dynamics [8]. Our design differs in two ways central to the research question: the "mean field" is a multidimensional, potentially *false* description, and units are born, die, and evolve. **Reflexivity and reactive measurement.** Self-fulfilling prophecy [5], reflexivity in markets [3], and the reactivity of rankings and metrics [4, 6] motivate our false-broadcast conditions; we contribute a platform where these effects are dosable and ablatable. **Signaling and the evolution of trust.** In sender–receiver games, receivers can evolve to ignore unreliable signals [9]. Our distrust result is consonant with that theory but arises in an open ecological population where the signal is *self-referential* — a compressed description of the receiving population itself — and where attention to it is a continuous heritable trait under selection from an energy economy, not a payoff matrix. **Agent-based modeling.** Our methodology follows the generative ABM tradition [10, 11]; the engine uses ideas from Mesa [12] and NetworkX [13]. Multi-agent reinforcement learning studies global state as an *architectural* aid [14]; we instead treat global information as an *experimental variable* with controlled veracity. Populations of language-model agents are now deployed with shared dashboards describing their own collective state [15]; the present rule-based setting provides causally clean baselines for that emerging regime.

3 The Model

3.1 Agents and ecology

Each agent i carries a heritable trait vector $\theta_i = (\mathrm{coop}, \mathrm{expl}, \mathrm{risk}, \mathrm{soc}, \mathrm{share}, \gamma_i) \in [0,1]^6$, where γ_i is its *global sensitivity* — how strongly broadcasts modulate its behavior. Agents hold energy; each step they pay metabolism and choose one action — harvest, share (costly transfer to a neighbor), connect, prune, reproduce, or idle — sampled with probabilities proportional to trait- and state-dependent propensities. The resource pool regenerates logistically, giving density-dependent competition. Reproduction requires an energy threshold and copies traits with Gaussian mutation ($\sigma = 0.05$); energy exhaustion or age causes death. Founding ages are staggered to avoid cohort artifacts.

3.2 Observer and broadcast conditions

Every $\Delta t = 10$ steps the observer computes the macrostate $S(t)$: population, fragmentation (1 minus largest-component share), degree- and betweenness-concentration, Freeman centralization [16], per-capita costly-helping rate, energy Gini, and turnover, plus mean trait values (logged, never broadcast). Five conditions define what agents receive (Table 1). Distorted broadcasts come in three modes: *invert* ($b_k = 1 - s_k$ per normalized metric), *crisis* (fixed alarming values), and *utopia* (fixed reassuring values). Broadcast values shift action propensities through fixed corrective response rules — e.g., reported fragmentation above 0.3 raises the propensity to form links; a reported cooperation deficit raises the propensity to share — each scaled by γ_i and a global gain g.

3.3 Reproducibility

The simulation is deterministic given (configuration, seed): one PRNG drives all choices, agents act in sorted order, and a state hash certifies replay. Identical hashes were verified across operating systems and

Table 1: Experimental conditions. All start from matched parameter distributions with paired random seeds.

Condition	Observer	Agents receive	Role
A — Local only	off	nothing	baseline
B — Observed, blind	on	nothing	controls for measurement
C — True feedback	on	accurate $S(t)$	main treatment
F — False feedback	on	distorted $S(t)$	reflexivity probe
N — Noise feedback	on	uniform random, same keys/ranges	stimulation control

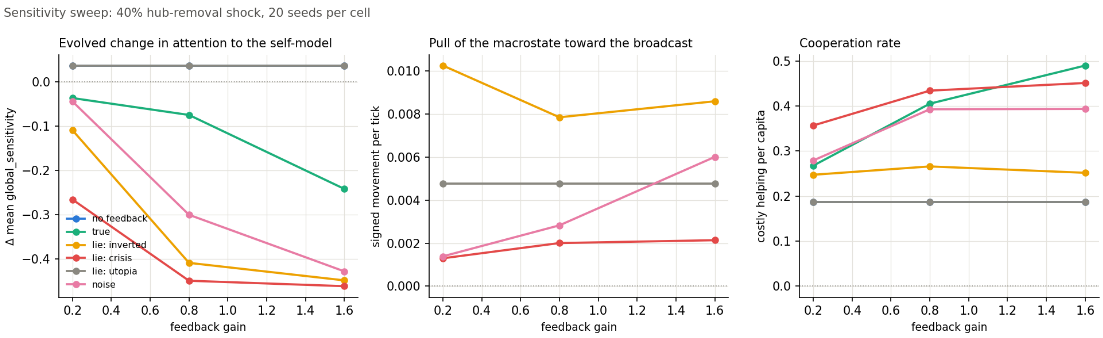

Figure 1: Sensitivity sweep (40% hub-removal shock; 20 seeds per cell; medians). Left: evolved change in mean global sensitivity γ — attention to the self-model is selected against in proportion to broadcast unreliability and gain; the flattering (utopia) lie tracks the no-feedback baseline exactly. Middle: story pull toward the broadcast. Right: cooperation rate.

machines. All code, configurations, and analysis scripts are public; every figure regenerates from two commands.

4 Experimental Design

Four outcomes were fixed before the first campaign: (1) post-shock recovery time (steps until population regains 90% of its pre-shock mean after standardized hub removal); (2) mean post-shock fragmentation; (3) per-capita costly-helping rate; (4) *story pull* — the mean signed movement of the macrostate toward the broadcast, counted only where broadcast and reality differ by more than 0.02, so that the measure is well defined under false and noise broadcasts. Campaigns: a *flagship* run (conditions A/B/C/$\mathrm{F_{invert}}$/N $\times$ 50 paired seeds, mild shock), a *harsh-shock* replication (removal of 40% of hubs, $\times$30 seeds), and a *sensitivity sweep* (feedback gain $g \in \{0.2, 0.8, 1.6\} \times \{\mathrm{A, C, F_{invert}, F_{crisis}, F_{utopia}, N}\} \times 20$ seeds): 970 runs of 4,000 steps. Group differences use Mann–Whitney U with Cliff's δ effect sizes; all other logged measures are exploratory.

5 Results

5.1 Measurement alone is causally inert

Under paired seeds, condition B is trajectory-identical to A: $\delta = 0.000$ on every outcome in both the flagship and harsh-shock campaigns. The instrument acquires causal power only when it broadcasts — a validation that the design isolates the channel of interest.

5.2 A false self-model pulls the macrostate toward itself

Story pull under the inverted false broadcast exceeds the noise control at every gain tested (flagship: median 0.0067 vs 0.0018, $\delta = 0.76$, $p \approx 6 \times 10^{-11}$; harsh shock: 0.0078 vs 0.0021, $\delta = 0.85$, $p \approx 1.6 \times 10^{-8}$). Relative to noise the effect is strongest at *low* gain (ratio ≈ 7 at $g = 0.2$): a quiet, systematic lie out-steers loud noise, while at high gain noise itself begins to steer (Fig. 1, middle). Spillovers are substantial: the inverted broadcast roughly halves post-shock fragmentation relative to baseline ($\delta = -0.50$, $p \approx 2 \times 10^{-5}$) because agents persistently told the network is fragmenting over-connect it.

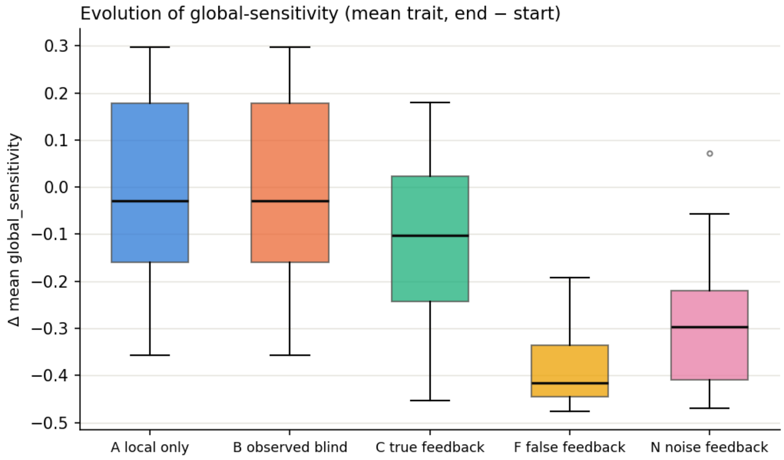

Figure 2: Evolution of attention to the self-model (harsh-shock campaign, $g = 0.8$, 30 seeds per condition). Change in population-mean global sensitivity $\bar{\gamma}$ between the start and end of the run. The ordering (false $<$ noise $<$ true $<$ none) emerged from selection alone.

5.3 The type of lie sets the direction

The three distortion modes produce qualitatively different regimes (Fig. 1). The *inverted* self-model, always reporting the mirror of reality, generates the strongest pull: it keeps every corrective response permanently active and thereby parasitizes the population's repair machinery. The *crisis* self-model is self-defeating for structure — corrections push fragmentation away from the alarmist report — yet its permanent cooperation-deficit signal mobilizes the highest costly-helping rates observed (0.36–0.45). The *utopia* self-model is behaviorally inert: because responses fire only on reported deficits, a flattering broadcast triggers nothing and is indistinguishable from disconnecting the channel, at every gain. Under purely corrective response rules, flattery equals silence.

5.4 Populations fed lies evolve to stop listening

Mean global sensitivity $\bar{\gamma}$ changes over $\sim$40 generations as an emergent, unprogrammed function of broadcast quality (Fig. 2). At $g = 0.8$ (harsh-shock campaign): no broadcast -0.03 (neutral drift), truth -0.10, noise -0.30 ($\delta = -0.77$), false -0.42 ($\delta = -0.94$, $p \approx 4 \times 10^{-10}$). The sweep reveals a dose–response: distrust of the false broadcast deepens from -0.11 ($g = 0.2$) to -0.45 ($g = 1.6$) — and at the highest gain even *accurate* feedback is selected against (-0.24): a broadcast that perturbs behavior strongly becomes a fitness liability even when true. The utopia lie, being consequence-free, breeds no distrust at all.

5.5 Accurate feedback as a distributed homeostat

True feedback roughly doubles costly helping (flagship: 0.397 vs 0.179, $\delta = 0.96$, $p \approx 10^{-16}$; replicated under harsh shock, $\delta = 0.95$), rising monotonically with gain ($0.27 \rightarrow 0.49$). The loop functions as a population-level thermostat: reported deficits are repaired by distributed individual action.

5.6 A robust negative result

Post-shock recovery time was insensitive to feedback in every condition and campaign, even with 40% of hubs removed (all $p > 0.12$; median ≈ 40 steps). In this ecology, resilience is demographic — fast regrowth — rather than informational. We report this as a boundary of the phenomenon rather than adjusting the protocol until the null disappears.

6 Discussion and Limitations

The first-order *directions* of response are encoded in the agents' corrective rules; that is by design, and we do not claim them as discoveries. The findings live one level up: the magnitude and asymmetry of steering (consistent lies $\gg$ matched noise, strongest at low gain), the spillovers to non-targeted structure, the qualitative taxonomy of lies (exploitative / mobilizing-but-self-defeating / inert), and above all the evolutionary layer, in which the credibility of a collective self-model becomes a selected property of the population. A conformist (non-corrective) response variant is a natural next ablation and we expect it to reverse parts of the taxonomy; the platform makes such tests one-line configuration changes. Other limitations: a single ecology and parameter family (though effects are monotone across an $8\times$ gain range), rule-based rather than learned agents, and one network per run rather than interacting populations.

The results have a practical reading for engineered systems. Metric-driven infrastructure — autoscalers, observability pipelines, trending algorithms — constitutes exactly this loop; the false-broadcast conditions model telemetry corruption, and the utopia result implies that *flattering telemetry is operationally equivalent to no telemetry* while raising no alarms. For multi-agent AI, where fleets of agents increasingly receive dashboards describing their own collective state, the evolution-of-distrust result suggests that persistent misreporting does not merely misdirect a population — it degrades the control channel itself.

7 Conclusion

We introduced a reproducible causal laboratory for reflexive self-modeling in adaptive networks and used it to show that (i) being measured changes nothing while being told changes much; (ii) a consistent false self-description steers a collective toward itself far more effectively than noise; (iii) the kind of lie determines whether it exploits, mobilizes, or pacifies; and (iv) selection acting on a heritable attention trait makes a population's trust in its own self-model an evolving quantity — one that unreliable observers spend down. Future work extends the platform to conformist response rules, continuity under total population turnover, and learned (including language-model) agents.

Reproducibility. Code, configurations, all 970 run manifests, and one-command reproduction of every figure: `github.com/Ivan825/Project-ONE`.

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